

**GE MEDICAL SYSTEMS****CT MANUFACTURING****DOCUMENT NUMBER 3476****REV. CxHino1****TITLE : HPower 60 System History Data****MFG #** _____**HHS DATA SHEET****GANTRY S/N** _____**TABLE S/N** _____**X-ray ``ON" Light**

X-ray ``ON" Lights Functional	Gantry Front - Yes/No	Gantry Back - Yes/No	Limit
			YES

TECH SIGNATURE _____ **DATE** _____**BOW Alignment**

High (mm)	Mid (mm)	Low (mm)	Limit
			0.7mm +/- 0.25mm

TECH SIGNATURE _____ **DATE** _____**Tomographic Plane**

External Measurement		Limit = I 1mm - S 1mm
Internal Measurement		Limit = I 1mm - S 1mm

TECH SIGNATURE _____ **DATE** _____**Exposure Time Accuracy**

Scan	Measured Time (seconds)	Limits
50 mm Scout Scan		0.56 - 0.60 sec.
150 mm Scout Scan		1.52 - 1.64 sec.
600 mm Scout Scan		5.84 - 6.32 sec.
0.8 sec. Axial Scan		0.768 - 0.832 sec.
1.0 sec. Axial Scan		0.960 - 1.04 sec.
4.0 sec. Axial Scan		3.84 - 4.16 sec.
30.1 sec. Helical Scan		28.9 - 31.3 sec.

TECH SIGNATURE _____ **DATE** _____**Collimator Aperture Check**

Aperture	Center Value	Limits
Small		3.783 +/- 5%
Large		5.9 +/- 5%

TECH SIGNATURE _____ **DATE** _____

**GE MEDICAL SYSTEMS****CT MANUFACTURING****DOCUMENT NUMBER 3476 REV. CxHino1****TITLE : HPower 60 System History Data****MFG # _____****HHS DATA SHEET****QA PHANTOM IMAGE QUALITY**

Resolution Quality - MTF		4 Slice Average		Limit
QA#1 / S / 10mm(2i) 120kV/260mA/1sec				.65 - 1.00
Contrast Scale		SL1	SL2	Lower Limit
QA#1 / S / 10mm(2i) 120kV/260mA/1sec				110.0 - 130.0

Visible Line Pattern	SL1	SL2	Limit
Series 2 Bone retro of above scans			B

Low Contrast Image Quality QA#2 / S / 10mm (2i) / 120KV / 260mA / 1sec

	SL1	SL3	SL5	SL7	Limits
Contrast					2.0 - 12.0
Holes Visible *					2-5

Low Contrast Image Specifications			
Contrast	2.00 to 3.99	4.00 to 7.99	8.00 to 12.00
Holes Visible *	2 - 5	3 - 5	4 - 5
Smallest Hole Size Visible	7.5 mm	5.0 mm	3.0 mm

* Number of holes required to be visible, depending on contrast factor.

2 of 4 slices must meet spec.

Image Noise Tests QA#3 / S / 10mm(2i) / 120KV / 260mA / 1sec

Means For Water		Means Limit	Standard Deviation For Water		Standard Deviation Limit
ROW 2A1A	ROW 1B2B		ROW 2A1A	ROW 1B2B	
		+/- 3.0			2.9 - 3.5

TECH SIGNATURE _____ **DATE** _____**Collimator Radiation Leakage Test**

Measurement (mR)		Limit = less than 1.5 mR
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MFG # _____ HHS DATA SHEET BLEEDER S/N _____

mA Meter Verification

Computer mA Hi (200mA) reading (mAHi). _____ Verify : | DVM – mAHi | <= 4mA _____

Computer mA Low (50mA) reading (mA Low). _____ Verify : | DVM – mA Low | <= 1mA _____

Note: DVM = DVM Reading / Resistance

Test Point MA_N to Test Point MA_GND

Computer mA Hi (200mA) reading (mAHi). _____ Verify : | DVM (x 100) – mAHi | <= 4mA _____

Computer mA Low (50mA) reading (mA Low). _____ Verify : | DVM (x 100) – mA Low | <= 1mA _____

TECH SIGNATURE _____ DATE _____

kV Meter Verification**Verification of HV Tank Feedback Resistors and Test Point KV_N to Test Point KV_GND**

Selected Technique	kV Measured at HV Divider	Divider kV Limit vs. Sel. kV	kV Measured at Test Points (See Note 1)	Test Points kV Limit vs. Sel. kV	Delta kV Divider (See Note 2)	Delta kV Test Point (See Note 3)	Delta kV Limit
80kV 50mA		+/- 2.4 KV		+/- 2.4 KV			+/- 1.6 kV
100kV 50mA		+/- 3.0 KV		+/- 3.0 KV			+/- 2.0 kV
120kV 50mA		+/- 3.6 KV		+/- 3.6 KV			+/- 2.4 kV
140kV 50mA		+/- 4.2 KV		+/- 4.2 KV			+/- 2.8 kV

Note 1: (kV_N to kV_GND) x 20

Note 3: Delta kV_Test Points = | Displayed – (Test Points) |

Note 2: Delta kV_Divider = | Displayed - Divider |

TECH SIGNATURE _____ DATE _____

Verification of Internal Scan Timer

Technic	Oscilloscope Exposure Duration (seconds)	Limit	Displayed Exposure Duration (seconds)	Limit
100 / 40		0.96 - 1.04 sec.		0.99 - 1.02 sec.

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GE MEDICAL SYSTEMS

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MFG # _____ LARGE FOCAL SPOT kV / mA HHS VERIFICATION DATA SHEET Use this data sheet if a printout cannot be obtained.

kV	mA	Time Sec	LSL	mA @ 5 mS (Scope)	USL	LSL 3.0%	mA (Console)	USL 3.0%	LSL 2.0%	kV (Scope)	USL 2.0%	LSL 2.0%	Time (Console)	USL 2.0%
80	10	1.00	8.0		12.0	9.7		10.3	78.4		81.6	0.980		1.020
80	20	1.00	16.5		23.5	19.4		20.6	78.4		81.6	0.980		1.020
80	50	1.00	42.0		58.0	48.5		51.5	78.4		81.6	0.980		1.020
80	100	1.00	84.5		115.5	97.0		103.0	78.4		81.6	0.980		1.020
80	200	1.00	169.5		230.5	194.0		206.0	78.4		81.6	0.980		1.020
80	400	1.00	339.5		460.5	388.0		412.0	78.4		81.6	0.980		1.020
100	10	1.00	8.0		12.0	9.7		10.3	98.0		102.0	0.980		1.020
100	20	1.00	16.5		23.5	19.4		20.6	98.0		102.0	0.980		1.020
100	50	1.00	42.0		58.0	48.5		51.5	98.0		102.0	0.980		1.020
100	100	1.00	84.5		115.5	97.0		103.0	98.0		102.0	0.980		1.020
100	200	1.00	169.5		230.5	194.0		206.0	98.0		102.0	0.980		1.020
100	400	1.00	339.5		460.5	388.0		412.0	98.0		102.0	0.980		1.020
100	420	1.00	356.5		483.5	407.4		432.6	98.0		102.0	0.980		1.020
120	10	1.00	8.0		12.0	9.7		10.3	117.6		122.4	0.980		1.020
120	20	1.00	16.5		23.5	19.4		20.6	117.6		122.4	0.980		1.020
120	50	1.00	42.0		58.0	48.5		51.5	117.6		122.4	0.980		1.020
120	100	1.00	84.5		115.5	97.0		103.0	117.6		122.4	0.980		1.020
120	200	1.00	169.5		230.5	194.0		206.0	117.6		122.4	0.980		1.020
120	400	1.00	339.5		460.5	388.0		412.0	117.6		122.4	0.980		1.020
120	440	1.00	373.5		506.5	426.8		453.2	117.6		122.4	0.980		1.020
140	10	1.00	8.0		12.0	9.7		10.3	137.2		142.8	0.980		1.020
140	20	1.00	16.5		23.5	19.4		20.6	137.2		142.8	0.980		1.020
140	50	1.00	42.0		58.0	48.5		51.5	137.2		142.8	0.980		1.020
140	100	1.00	84.5		115.5	97.0		103.0	137.2		142.8	0.980		1.020
140	200	1.00	169.5		230.5	194.0		206.0	137.2		142.8	0.980		1.020
140	380	1.00	322.5		437.5	368.6		391.4	137.2		142.8	0.980		1.020

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Author: Hirofumi Yanagita

Date: Revised 12/3 03 by Hino CT Eng.

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MFG # _____ SMALL FOCAL SPOT kV / mA HHS VERIFICATION DATA SHEET Use this data sheet if a printout cannot be obtained.

kV	mA	Time Sec	LSL	mA @ 5 mS (Scope)	USL	LSL 3.0%	mA (Console)	USL 3.0%	LSL 2.0%	kV (Scope)	USL 2.0%	LSL 2.0%	Time (Console)	USL 2.0%
80	10	1.00	8.0		12.0	9.7		10.3	78.4		81.6	0.980		1.020
80	20	1.00	16.5		23.5	19.4		20.6	78.4		81.6	0.980		1.020
80	50	1.00	42.0		58.0	48.5		51.5	78.4		81.6	0.980		1.020
80	100	1.00	84.5		115.5	97.0		103.0	78.4		81.6	0.980		1.020
80	200	1.00	169.5		230.5	194.0		206.0	78.4		81.6	0.980		1.020
80	300	1.00	254.5		345.5	291.0		309.0	78.4		81.6	0.980		1.020
100	10	1.00	8.0		12.0	9.7		10.3	98.0		102.0	0.980		1.020
100	20	1.00	16.5		23.5	19.4		20.6	98.0		102.0	0.980		1.020
100	50	1.00	42.0		58.0	48.5		51.5	98.0		102.0	0.980		1.020
100	100	1.00	84.5		115.5	97.0		103.0	98.0		102.0	0.980		1.020
100	200	1.00	169.5		230.5	194.0		206.0	98.0		102.0	0.980		1.020
120	10	1.00	8.0		12.0	9.7		10.3	117.6		122.4	0.980		1.020
120	20	1.00	16.5		23.5	19.4		20.6	117.6		122.4	0.980		1.020
120	50	1.00	42.0		58.0	48.5		51.5	117.6		122.4	0.980		1.020
120	100	1.00	84.5		115.5	97.0		103.0	117.6		122.4	0.980		1.020
120	200	1.00	169.5		230.5	194.0		206.0	117.6		122.4	0.980		1.020
140	10	1.00	8.0		12.0	9.7		10.3	137.2		142.8	0.980		1.020
140	20	1.00	16.5		23.5	19.4		20.6	137.2		142.8	0.980		1.020
140	50	1.00	42.0		58.0	48.5		51.5	137.2		142.8	0.980		1.020
140	100	1.00	84.5		115.5	97.0		103.0	137.2		142.8	0.980		1.020
140	170	1.00	144.0		196.0	164.9		175.1	137.2		142.8	0.980		1.020

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